



Cyber-Physical Systems
in Mechanical Engineering TU Berlin

PyTorch Transformer

TU Berlin Summer School 2026 – DL4TS

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Dataset and Task



Dataset:

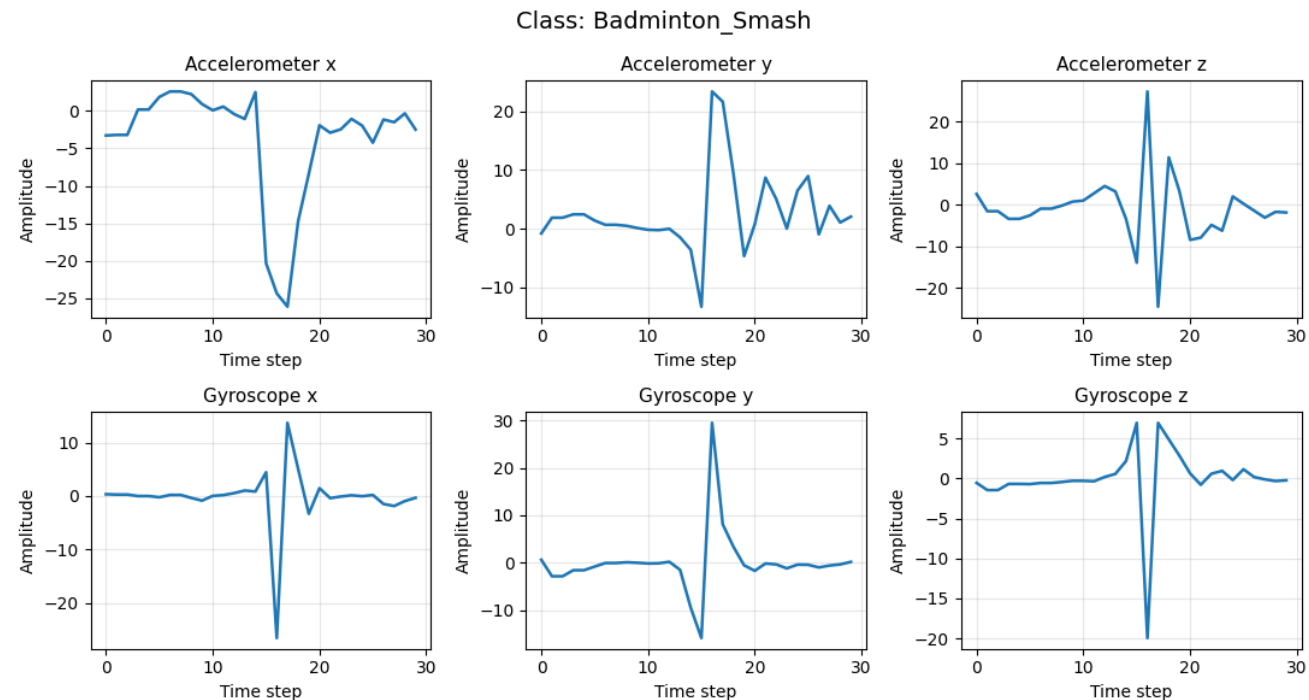
- RacketSport multivariate time-series classification dataset
- Each sample: 30 time steps with 6 input features $X \in \mathbb{R}^{(30 \times 6)}$
- 4 classes: ['Badminton_Clear', 'Badminton_Smash', 'Squash_Backhand', 'Squash_Forehand']

Classification task:

- Input: Multivariate sequence X
- Output: class label y

Implementation:

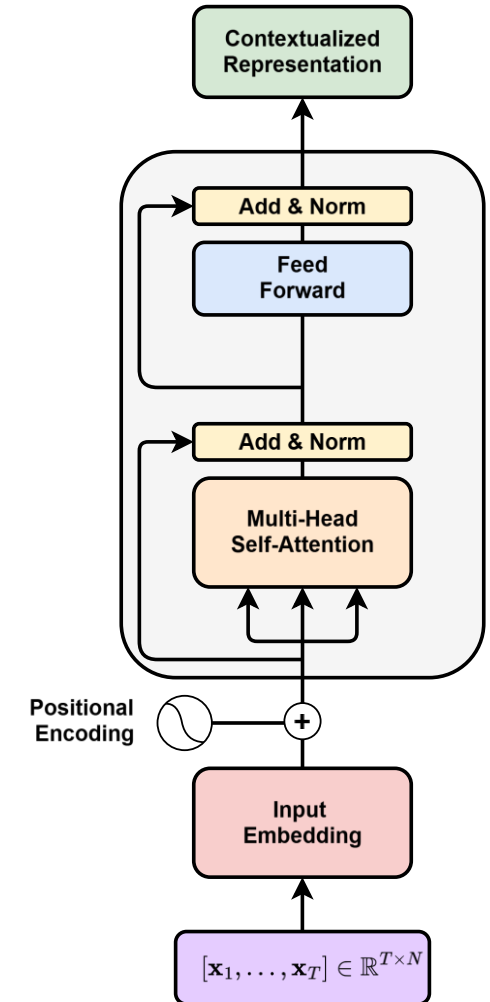
- Encoder-Only Transformer
- CLS-Token
- Validation workflow



Encoder-Only Transformer



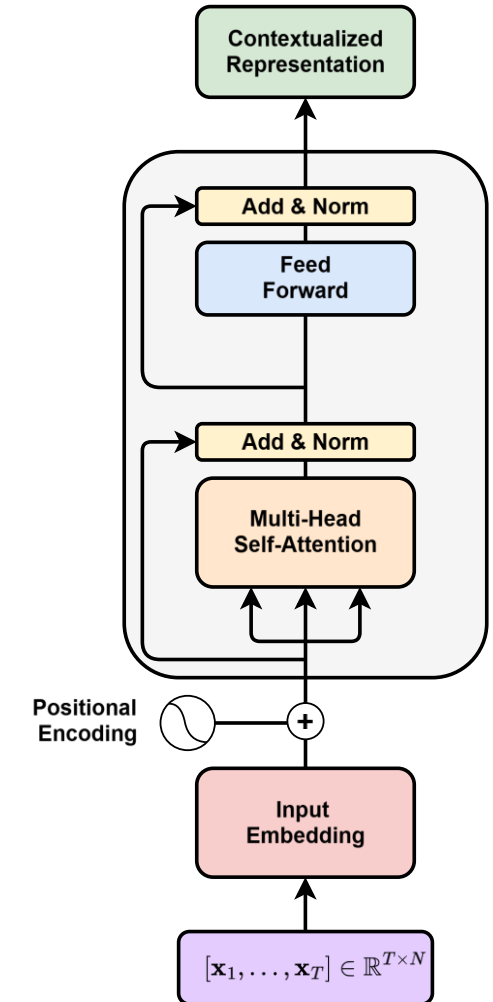
- **Input:** $X = [x_1, \dots, x_T] \in \mathbb{R}^{(T \times N)}$
- **Project to model dimension:** $X \in \mathbb{R}^{(T \times d_{\text{model}})}$
 - Map the feature dimension from N to d_{model}
 - e.g. via a linear projection
- **Contextualized representation:** $Z \in \mathbb{R}^{(T \times d_{\text{model}})}$
- **Task-specific processing**
 - Seq2Seq regression
 - Classification
 - Forecasting



Time Series Classification



- **Contextualized representation:** $Z \in \mathbb{R}^{(T \times d_{\text{model}})}$
- 1. Map the contextualized representation to a scalar output**
 - Flatten $Z \in \mathbb{R}^{(T \times d_{\text{model}})} \rightarrow \text{zflat} \in \mathbb{R}^{(T * d_{\text{model}} \times 1)}$
 - e.g. $T = 500$, $d_{\text{model}} = 64 \rightarrow \underline{32001}$ trainable parameters
 - 2. Mean or max pooling**
 - $z_{\text{mean}} = \frac{1}{T} \sum_{t=1}^T z_t \in \mathbb{R}^{d_{\text{model}}}$ or $z_{\text{max}} \in \mathbb{R}^{d_{\text{model}}}$
 - e.g. $T = 500$, $d_{\text{model}} = 64 \rightarrow \underline{65}$ trainable parameters
 - 3. CLS Token**
 - Append a CLS Token to the start of the sequence
 - e.g. $T = 500$, $d_{\text{model}} = 64 \rightarrow \underline{64 + 65 = 129}$ trainable parameters



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